

Finance REM-Card Capture and Reconciliation

A portfolio-safe reconstruction of a casino finance operation where handwritten physical documents were converted into structured, auditable data using AI Builder, Azure AI Foundry, Azure Functions, Power Automate, and Power BI.

Sanitized implementation case study

Physical

Handwritten REM cards

Structured

Tabular finance data

Rule-based

Variance checks

Audit-ready

Reconciliation output

What this shows: Finance operations depended on handwritten REM cards and related operational records. Important values were written on paper and needed to be captured into a consistent table before reports could be reconciled. The challenge was not only reading the document; it was preserving the relationship between card, account, balances, payments, vouchers, VAT, and gaming adjustments.

End-to-end workflow

1

Paper REM card

Source enters the process.

2

Capture + review

Context and missing fields are identified.

3

Structured table

Data becomes consistent and usable.

4

Reconcile rules

Rules compare, route, or reconcile.

5

Variance report

People receive an action-ready result.

Business value

- Transforms unstructured or incomplete source information into consistent records.
- Separates automation from human review so exceptions are visible.
- Creates a traceable path from original source to report or action.
- Supports operational teams with clearer ownership and faster review.

How to read this project

HR can review the first two pages for the business process and my contribution. The linked repository folder contains sanitized data models, workflow examples, documentation, and implementation notes for deeper technical review.

How the solution works

The important part of this project is the complete operating chain: source capture, structured transformation, business-rule processing, exception handling, and reporting. Each stage produces a clear output that can be validated before the next stage uses it.

01 / DOCUMENT INTAKE	Physical REM cards and supporting documents were collected as the source of truth. Each image or scan was associated with a processing batch and a business date so the resulting record could be traced back to the original document. Output: traceable, reviewable data for the next stage
02 / FIELD CAPTURE	The extraction process captured REM card number, account number, REM card balance, cash issued, chip payments, cash payments, and other transaction fields. Because handwriting and layouts can vary, captured values required validation and exception review. Output: traceable, reviewable data for the next stage
03 / TABULAR STAGING	The document data was normalized into a structured table with consistent field names, data types, dates, amounts, and reference identifiers. This staging layer made the records usable in Excel, Power Query, Power Automate, SQL, or Dataverse-style storage. Output: traceable, reviewable data for the next stage
04 / RECONCILIATION	Operational totals and financial movements were compared across source records and reports. The reconciliation logic covered vouchers in, vouchers out, VAT in, VAT out, free bets, match play logs, cash movements, chip movements, and balance differences. Output: traceable, reviewable data for the next stage
05 / EXCEPTION REPORTING	Variances were isolated instead of being hidden inside totals. Exceptions could then be reviewed by finance operations with the original document reference, expected amount, captured amount, variance, and review status. Output: traceable, reviewable data for the next stage

Example structured record

```
{ "source_reference": "BATCH-2026-001 / DOCUMENT-0042", "record_id": "REC-0042", "status": "review_required", "captured_fields": { "account_number": "sample-account", "document_number": "sample-document", "amount": 0, "tax": 0 }, "checks": { "required_fields": "passed", "duplicate": "not_found", "reconciliation": "variance_found" }, "next_action": "finance_review" }
```

Design principle: Never hide an exception inside a final total. Keep the source reference, extracted value, expected value, variance, and review status together so an operations user can explain what happened.

Data and control points

Control point	Purpose	Typical output
Source reference	Connects the structured row to the physical or digital document.	Batch ID, file ID, page or document reference
Validation	Prevents missing, malformed, duplicate, or low-confidence data from flowing silently.	Pass, fail, review required
Business rules	Applies finance or service logic to compare, route, calculate, or approve.	Matched, variance, assigned queue
Exception record	Makes human review explicit and measurable.	Expected, actual, variance, owner, status
Reporting layer	Turns operational records into trends and actions.	Volume, aging, variance, SLA, resolution

Responsibilities and technology map

My contribution

- Mapped the operational process from source document or user request to final report.
- Defined fields, validation checks, exception states, and review ownership.
- Designed automation steps and structured payloads for repeatable processing.
- Connected data preparation with reporting and operational follow-up.
- Documented the workflow so another team member can understand, test, and operate it.

Portfolio boundaries

This is a sanitized portfolio representation. Casino/client names, real account numbers, production URLs, credentials, and confidential records are intentionally excluded. The domain workflow and technical responsibilities are preserved at a useful level for discussion.

Capability matrix

Capability	Evidence in this project	Keywords
Process automation	Multi-stage intake, transformation, validation, routing, exception, and reporting flow.	Power Automate, workflow design, integration
Data engineering	Structured fields, normalization, staging tables, checks, and downstream reporting data.	Python, SQL, Excel, Power Query, Dataverse
Finance operations	Document capture, transaction fields, reconciliation, variance identification, review states.	OCR, reconciliation, controls, audit trail
Conversational / AI patterns	Where applicable, intent, confidence, fallback, and human-review concepts.	Copilot Studio, Azure AI, Azure OpenAI
Reporting	Operational metrics, variance categories, aging, SLA, resolution and trend views.	Power BI, Microsoft Fabric, dashboards

Technology stack

OCR / document capture

AI Builder

Azure AI Foundry

Azure Functions

Python

Power Automate

Excel

Power Query

SQL

Dataverse

Power BI

App Insights

Validation rules

Variance reporting

Audit trail

REST APIs

Git / CI/CD

Suggested interview walkthrough: Start with the source document or request, explain how you preserved traceability, show the validation and reconciliation decision points, then explain how the final report helped an operations or support user take action.